

CO3 AT3 MOCK INTERVIEW

Student 28 **Sivarangjenii B** 192521375RD

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Q1. DICTIONARY

ANSWER

A dictionary can be used as a contact book by storing the contact name as the key and the phone number as the value. it allows contacts to be added, updated, searched, and deleted easily.
example: {"ranju": "4842498", "Rangju": "4577842"}

Q2. LIST

ANSWER

list comprehension provides a short way to create a new list from an existing list. it can also apply conditions while creating the list.
example: even numbers can be replaced with 0 while odd numbers remain unchanged
example input and output : i/p: [1,2,3] o/p: [1,0,3]

Q3. SET

ANSWER

A "subset" is a set in which every element is present in another set ,
a "superset" is a set that contains all the elements of another set
example: if $A = \{1,2\}$ and $B = \{1,2,3\}$
then A is a subset of B
and B is a superset of A

Q4. DICTIONARY

ANSWER

A Dictionary is faster for searching because it uses hashing ,
While a list searches elements one by one.
List search : $O(n)$ time.
A dictionary uses hashing so searching for a key takes $O(1)$ average time
Therefore a dictionary is generally faster than a list when frequent searching is required

Q5. LIST

ANSWER

Both are used to create values using a compact expression . List comprehension creates the entire list immediately and stores all values in memory,
While a generator expressions produces values one at a time, using less memory
for example: list comprehension uses $x*x$ for x in $\text{range}(5)$,
whereas a generator uses $x*x$ for x in $\text{range}(5)$